Yashvant A T K

Software Engineer | Full-Stack Development | Agentic AI & LLM Applications | DSA
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Summary

CS undergraduate (CGPA 9.25) who builds production-style software end to end: **React + TypeScript** front-ends, **FastAPI / Node.js** back-ends with REST APIs, and **LLM agents** with tool calling and **MCP**, shipped with **Docker** and **GitHub Actions CI**. Focused on AI systems whose output is verified before anyone relies on it. 2x national hackathon winner and ICPC India 2026 competitor.

Education

Indian Institute of Information Technology, Design and Manufacturing — Chennai, India
B.Tech in Computer Science and Engineering | *CGPA: 9.25/10.0* — *Expected May 2029*

- **Coursework:** Data Structures, Design and Analysis of Algorithms, Object-Oriented Programming, Database Systems, Information Systems.

Projects

AutoResolve – Multi-Agent Code Review & Repair | *Rust, TypeScript, Node.js, MCP, Docker* | Code — Sep 2026

- A deterministic Rust controller drives **7 specialised LLM agents**; a fix is kept only if a regression test **fails before and passes after** the patch, inside an isolated **Docker** sandbox.
- Ships as a TypeScript CLI (Zod, ts-morph AST), **VS Code extension**, **MCP server** (JSON-RPC 2.0) and **GitHub Action**; 22-case benchmark: 100% single-bug detection, 85% multi-bug recall, 0 false alarms.

AURA AI – Smart Traffic Digital Twin | *React 19, Vite, FastAPI, Pydantic, C++, YOLOv8* | Live | Code — Aug 2026

- Full-stack platform of 4 services: YOLOv8 video ingestion, a **C++ max-flow optimisation engine**, a typed **FastAPI + Pydantic** REST back-end, and a **React 19 + React Router 7** dashboard.
- Deployed the API with Uvicorn/Gunicorn on **Render** and the front-end on **Vercel**.

CiteScout AI – Agentic Literature Scout | *Python, Gemini Function Calling, async httpx, SQLite* | Code — Oct 2026

- Tool-calling agent that searches **5 scholarly APIs**, merges duplicates and verifies every citation (DOI, title, authors, year) through validation gates and human approval before output.
- Async networking with retries, exponential backoff and a SQLite cache; automated tests in **GitHub Actions**. Team lead, Ziro Agentic AI Hackathon 2026.

Nexus Learning – AI Syllabus Orchestrator | *Python, Streamlit, Gemini API* | Live | Code — Mar 2026

- Turns syllabus documents into structured study plans using schema-validated XML/JSON LLM output; **1st place at Vashisht '26** among 300+ teams, built by my 4-person team in 36 hours.

Developer Portfolio | *React, TypeScript, Vite, Tailwind CSS v4, Framer Motion* | Live | Code — 2026

- Responsive animated site with client-side routing, typed reusable components and continuous deployment on Vercel.

CP-Team-Tracker | *FastAPI, SQLite, JavaScript* | Code — Jul 2026

- REST API with OpenAPI/Swagger docs over a normalised SQLite schema, plus a decoupled JS dashboard and one-command setup scripts.

Experience

Mars Rover Society (Ares-Nova), IIITDM Kancheepuram — Chennai, India
Software Wing Member — *Apr 2026 – Present*

- Develop and debug concurrent C++ and Python modules for the rover's ROS2 software stack in a team with code reviews and shared documentation.

Technical Affairs Team, Student Affairs Council, IIITDM Kancheepuram — Chennai, India
Manager & Core Organiser, Savara Fest — *Jul 2026 – Present*

- Led a cross-functional volunteer team to run **Savara Fest**, the institute's technical festival, owning logistics and hackathon workflows; turn requests from student leads and faculty into milestones and tracked deliverables.

CS Club – AI Wing, IIITDM Kancheepuram — Chennai, India
Core Member — *Aug 2026 – Present*

- Run workshops and peer study groups on LLM tools and agentic AI for the campus developer community.

Achievements

Decode SIH Hackathon – National 1st Place & Team Lead among 3,500+ teams (2026) | **Vashisht '26 – 1st Place & Team Lead** (Mar 2026).

ICPC India 2026 participant | **JEE Advanced 2025** AIR 11,254 | **NSEP 2024** State Top 1%.

Technical Skills

Languages: Python, C++, TypeScript, JavaScript (ES6+), SQL, Rust, C, HTML/CSS

Front-End: React 19, React Router, Vite, Tailwind CSS, Framer Motion

Back-End & Data: FastAPI, Pydantic, Node.js, Express.js, RESTful APIs, WebSockets, Zod, MySQL, SQLite

AI / LLM: Gemini, OpenAI & Claude APIs, Function Calling, Model Context Protocol (MCP), Agentic Workflows, Prompt Engineering, YOLOv8

DevOps & Tools: Docker, GitHub Actions (CI/CD), Git, Linux, Vercel, Render, ROS2, Streamlit

Fundamentals: Data Structures, Algorithms, OOP, DBMS, Concurrency, Testing & Debugging, Agile